AP PRECALCULUS

Unit 1 • Lesson 1.5 notes

Lesson 1.5 • Polynomial Functions and Complex Zeros

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Connect real zeros, factors and x-intercepts.
- Use multiplicity to describe the graph at each zero.
- Use the Fundamental Theorem of Algebra and conjugate pairs; test for even and odd functions.

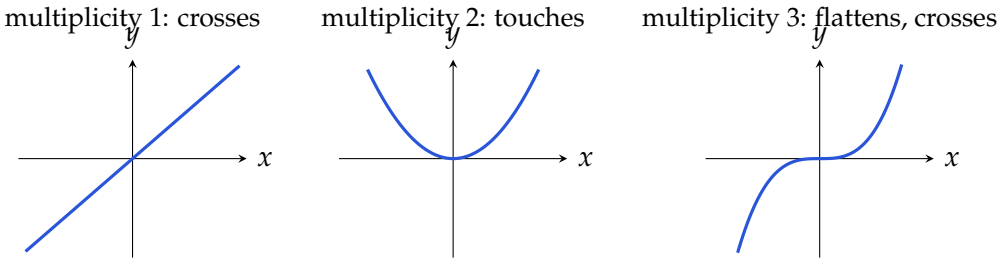
Vocabulary. zero, factor, multiplicity, complex number, conjugate, even function, odd function.

1. Zeros, factors, intercepts

For a polynomial p, the following say the same thing: a is a real zero; $(x-a)$ is a factor of $p(x)$; $(a,0)$ is an x-intercept of the graph.

2. Multiplicity

If $(x-a)^k$ is a factor and $(x-a)^{k+1}$ is not, a has multiplicity k.



3. Complex zeros

Key idea

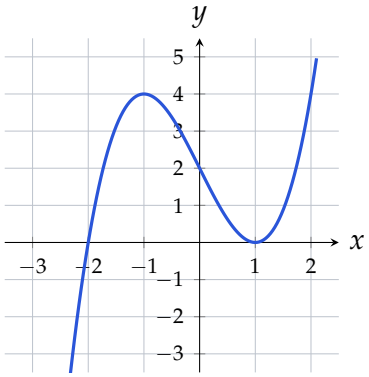
A polynomial of degree n has exactly n complex zeros, counted with multiplicity. If its coefficients are real, non-real zeros come in conjugate pairs $a \pm bi$, so the number of real zeros has the same parity as n (a cubic always has at least one real zero).

Example 1 • Building a polynomial

Write a polynomial of least degree with real coefficients, leading coefficient 1, zeros -1 (multiplicity 2) and $2i$.

Solution. $-2i$ must also be a zero: $p(x) = (x+1)^2(x-2i)(x+2i) = (x+1)^2(x^2+4)$, degree 4.

Example 2 • Graph to formula



Solution. The graph crosses at $x = -2$ and touches at $x = 1$, and rises to the right. A cubic with these features: $q(x) = a(x+2)(x-1)^2$ with $a > 0$. The y-intercept is 2, so $a \cdot 2 \cdot 1 = 2$ and $a = 1$.

Even and odd functions

Even: $f(-x) = f(x)$ (symmetric about the y-axis), for example $x^4 - 3x^2$. Odd: $f(-x) = -f(x)$ (symmetric about the origin), for example $x^3 - 5x$. A polynomial with only even powers is even; with only odd powers, odd.

Common mistakes

- Listing $\pm 2i$ as x-intercepts. Non-real zeros do not appear on the graph.
- Writing the factor for the zero -3 as $(x-3)$ instead of $(x+3)$.

Quick check

1. How many real zeros can a polynomial of degree 4 with real coefficients have?
2. Is $f(x) = x^5 - 2x^3 + x$ even, odd, or neither?

Answers. 1. 0, 2 or 4 (counted with multiplicity); never 1 or 3. 2. Odd (only odd powers).